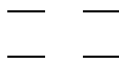


1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned} a_u^l &= +b_{1g} + a_u \\ b_{1u}^l &= +a_g + b_{1u} \\ b_{2g}^l &= +b_{3u} + b_{2g} \\ b_{3g}^l &= +b_{2u} + b_{3g} \\ a_u^r &= -b_{1g} + a_u \\ b_{1u}^r &= -a_g + b_{1u} \\ b_{2g}^r &= -b_{3u} + b_{2g} \\ b_{3g}^r &= -b_{2u} + b_{3g} \end{aligned}$$

2 Monomer States and Configurations

$$\begin{aligned}
S &= + \begin{pmatrix} \overline{} & \overline{} \\ \downarrow\uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
Q &= + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\
i^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{2u}} \right| - \left| \underbrace{(b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{2u}} \right| \\
e^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{2u}} \right| + \left| \underbrace{(b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{2u}} \right| \\
e^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{3u}} \right| - \left| \underbrace{(b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{3u}} \right| \\
i^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{3u}} \right| + \left| \underbrace{(b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{3u}} \right|
\end{aligned}$$

3 Linear Combinations

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \end{pmatrix} + \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{a_g^r} \right|$$

summed up:

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$-2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right|$$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\downarrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \downarrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\downarrow\uparrow} & \overline{\downarrow\uparrow} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right|$$

$$- \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{a_g^r} \right|$$

summed up:

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right|$$

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right|$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right|$$

substitution by dimer:

$$\begin{aligned} & + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \end{aligned}$$

summed up:

$$\begin{aligned} & + \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \end{aligned}$$

$$\begin{aligned} & + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| \end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\ + \quad 2 \quad \cdot \quad \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \quad 2 \quad \cdot \quad \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|$$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| \\ +4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| \\ -4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{2u}^l} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right|$$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+ \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|$$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{2u}^l} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|$$

summed up:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right|$$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|$$

substitution by dimer:

$$\begin{aligned} & + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \end{aligned}$$

summed up:

$$\begin{aligned} & + \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \end{aligned}$$

$$\begin{aligned} & + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|$$

summed up:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\ + \quad 2 \quad \cdot \quad \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| \quad - \quad 2 \quad \cdot \quad \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

monomer ci vectors:

substitution by dimer:

summed up:

$$\begin{array}{l}
+4 \cdot \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} - 4 \cdot \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
+4 \cdot \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} + 4 \cdot \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
-4 \cdot \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} - 4 \cdot \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}}
\end{array}$$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|$$

summed up:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

$$+ \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_g
$S * Q - Q * S$	b_{1u}
$i^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$	b_{1u}
$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$	a_u
$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$	a_u
$e^3 b_{2u} * e^3 b_{2u}$	a_g
$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$	b_{1u}
$i^3 b_{3u} * i^3 b_{3u}$	a_g

$\Rightarrow$ 7x a_g and 3x b_{1u} and 4x b_{1g} and 4x a_u

5 CI Vector Conclusion

Dimer State	CI Vectors
$S * Q + Q * S$	$+2 \cdot 0220aaaa + 2 \cdot 02aaaa20 - 2 \cdot 0a2aa2a0 + 2 \cdot 0aa0a22a $ $+2 \cdot a22a0aa0 - 2 \cdot a2a00a2a + 2 \cdot aa2002aa + 2 \cdot aaaa0220 $
$S * Q - Q * S$	$+2 \cdot 022aaaa0 - 2 \cdot 02a0aa2a + 2 \cdot 0a20a2aa + 2 \cdot 0aaaa220 $ $-2 \cdot a2200aaa - 2 \cdot a2aa0a20 + 2 \cdot aa2a02a0 - 2 \cdot aaa0022a $
$i^3 b_{2u} * i^3 b_{2u}$	$+ a2a0a2a0 - 2 \cdot a2a00a2a - 2 \cdot a22a0aa0 + 2 \cdot aaaa0220 $ $-2 \cdot aa2002aa - 2 \cdot 02aaaa20 + 2 \cdot 0220aaaa - 2 \cdot 0aa0a22a $ $-2 \cdot 0a2aa2a0 + 0a2a0a2a $
$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$	$+2 \cdot a2a0a2a0 - 2 \cdot 0a2a0a2a $
$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$	$+4 \cdot a2aa0a20 - 4 \cdot a2200aaa + 4 \cdot aaa0022a + 4 \cdot aa2a02a0 $ $-4 \cdot 02a0aa2a - 4 \cdot 022aaaa0 + 4 \cdot 0aaaa220 - 4 \cdot 0a20a2aa $
$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$	$+2 \cdot a2a0aa20 + 2 \cdot aa20a2a0 - 2 \cdot a2a002aa - 2 \cdot 02aaa2a0 $ $-2 \cdot 0a2aaa20 - 2 \cdot aa200a2a + 2 \cdot 0a2a02aa + 2 \cdot 02aa0a2a $
$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$	$-2 \cdot a220aaa0 + 2 \cdot aaa0a220 + 2 \cdot a2aa02a0 - 2 \cdot 02a0a2aa $ $+2 \cdot 0a20aa2a - 2 \cdot aa2a0a20 - 2 \cdot 0aaa022a + 2 \cdot 022a0aaa $
$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$	$+2 \cdot a2a0aa20 + 2 \cdot aa20a2a0 + 2 \cdot a2a002aa + 2 \cdot 02aaa2a0 $ $-2 \cdot 0a2aaa20 - 2 \cdot aa200a2a - 2 \cdot 0a2a02aa - 2 \cdot 02aa0a2a $
$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$	$-2 \cdot a220aaa0 + 2 \cdot aaa0a220 - 2 \cdot a2aa02a0 + 2 \cdot 02a0a2aa $ $+2 \cdot 0a20aa2a - 2 \cdot aa2a0a20 + 2 \cdot 0aaa022a - 2 \cdot 022a0aaa $
$e^3 b_{2u} * e^3 b_{2u}$	$+ a2a0a2a0 + 2 \cdot a2a00a2a + 2 \cdot a22a0aa0 - 2 \cdot aaaa0220 $ $+2 \cdot aa2002aa + 2 \cdot 02aaaa20 - 2 \cdot 0220aaaa + 2 \cdot 0aa0a22a $ $+2 \cdot 0a2aa2a0 + 0a2a0a2a $
$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$	$+2 \cdot a2a0aa20 + 2 \cdot aa20a2a0 - 2 \cdot a2a002aa - 2 \cdot 02aaa2a0 $ $+2 \cdot 0a2aaa20 + 2 \cdot aa200a2a - 2 \cdot 0a2a02aa - 2 \cdot 02aa0a2a $
$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$	$-2 \cdot a220aaa0 + 2 \cdot aaa0a220 + 2 \cdot a2aa02a0 - 2 \cdot 02a0a2aa $ $-2 \cdot 0a20aa2a + 2 \cdot aa2a0a20 + 2 \cdot 0aaa022a - 2 \cdot 022a0aaa $
$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$	$+2 \cdot a2a0aa20 + 2 \cdot aa20a2a0 + 2 \cdot a2a002aa + 2 \cdot 02aaa2a0 $ $+2 \cdot 0a2aaa20 + 2 \cdot aa200a2a + 2 \cdot 0a2a02aa + 2 \cdot 02aa0a2a $
$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$	$-2 \cdot a220aaa0 + 2 \cdot aaa0a220 - 2 \cdot a2aa02a0 + 2 \cdot 02a0a2aa $ $+2 \cdot 0a20aa2a - 2 \cdot aa2a0a20 + 2 \cdot 0aaa022a - 2 \cdot 022a0aaa $